

Shubham Rothe

Data Engineer — Databricks — PySpark — Python — SQL — Cloud — ETL
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PROFESSIONAL SUMMARY

Data Engineer with **4 years of experience** designing, developing, and supporting scalable **ETL/ELT pipelines** for financial, banking, and loan datasets. Strong hands-on expertise in **Databricks, PySpark, SQL Server (T-SQL), MySQL Workbench, Python, and Prophecy ETL**, delivering cloud-native solutions using **Lakehouse and Medallion Architecture**. Experienced in data ingestion, transformation, orchestration, validation, and production support.

Skilled in **Apache Spark, Snowflake, AWS, Azure, batch and streaming data processing, SCD Type-2 modeling, and data pipeline optimization**. Proven ability to build **scalable data pipelines, data quality frameworks, and reliable analytics systems** with a focus on performance and data integrity. Interested in working with **Azure Data Factory, Apache Airflow, and Kafka**, with a good conceptual understanding.

Skills

Languages: Python, SQL/T-SQL

Big Data: Apache Spark, PySpark, Delta Lake, ETL/ELT pipelines

Cloud and Data Platforms: Databricks, Delta Lake, Medallion Architecture, Snowflake, AWS, Azure

Databases: MS SQL Server (SSMS), MySQL Workbench, MongoDB

DevOps: GitLab, JAMS Scheduler, Terraform (IaC), Monitoring and Observability

ETL & Orchestration: Prophecy, Pentaho

Core Competencies: Data Modeling, Medallion Architecture, Batch Processing, Stream Processing, Data Warehousing, Data Lakehouse, Data Governance, Unit Testing, OOP, Data Structures and Algorithms, Star Schema, Snowflake Schema

Interested to work in: Apache Kafka, Apache Airflow, dbt, Azure Data Factory, Azure Synapse

Work Experience

Software Development Engineer – Data Engineer

Decimal Point Analytics, Mumbai (HQ) — Working Location: Nashik

Jul 2022 – Present

Project: Servicer Data Onboarding (Medallion Architecture)

- Led and coordinated a team of 2 junior developers and conducted a **20-day structured training program** on **PySpark, SQL, and Databricks** for newly hired engineers, accelerating onboarding and reducing ramp-up time.
- Designed and developed scalable ETL/ELT pipelines in Databricks and Prophecy to process financial datasets of **10M+ records** from CSV, Excel, JSON, and text files.
- Implemented Medallion Architecture (Bronze–Silver–Gold) pipelines enabling raw ingestion, cleansing, validation, and analytics-ready datasets.
- Automated file ingestion from FTP to AWS S3 using Python, followed by Databricks-based transformations.
- Developed parameterized and reusable Databricks notebooks for ingestion, transformation, validation, and reporting.
- Implemented Delta Lake features including **MERGE, OPTIMIZE, Z-ORDER, VACUUM**, and schema evolution.
- Built data quality and validation frameworks with logging and audit controls.
- Orchestrated end-to-end workflows using **JAMS Scheduler** with failure handling and SLA monitoring.
- Implemented a **parallel execution framework** in Databricks using a controller notebook to dynamically trigger ingestion and transformation notebooks concurrently across multiple deals and files, eliminating sequential bottlenecks and improving pipeline throughput.
- Leveraged **Databricks Auto Loader (cloudFiles)** with structured streaming, schema inference, and **checkpointing** to enable scalable, fault-tolerant incremental ingestion.
- Optimized Spark jobs using partitioning, caching, broadcast joins, and query tuning, improving performance by up to **40%**.
- Migrated **100+ Databricks jobs** from DBFS/mount paths to **Unity Catalog external volumes**.
- Provided production support, incident troubleshooting, and release validation.

Project: US Banking & Client Data Integration

- Developed dynamic stored procedures, functions, and parameter-driven SQL logic for financial reporting.
- Applied complex business rules across fact and dimension tables using optimized joins and aggregations.
- Automated incremental and quarterly data loads with accuracy and consistency checks.
- Designed ER diagrams and optimized database schemas for performance and scalability.

Project: Mutual Fund Automation

- Developed Pentaho ETL workflows for NAV and holdings data ingestion from CSV, text, and Excel files.
- Designed fact and dimension tables using star schema modeling.
- Implemented incremental loads and validation checks that **achieved 99.8% data accuracy**.

Education

Bachelor of Engineering (Information Technology)

Savitribai Phule Pune University

CGPA: 9.03

2018 – 2022